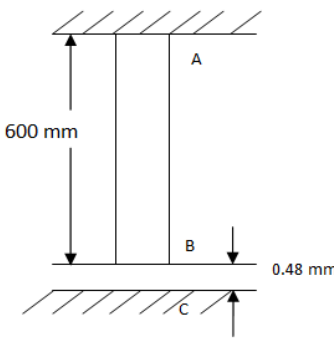
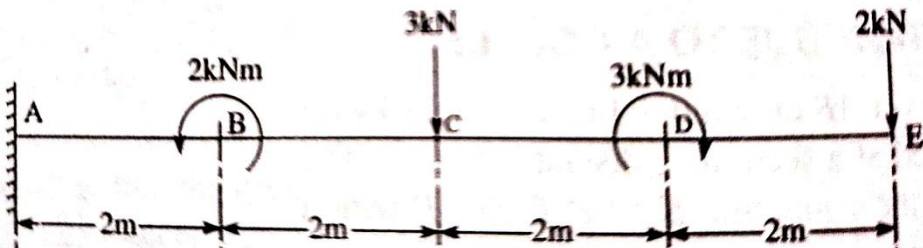
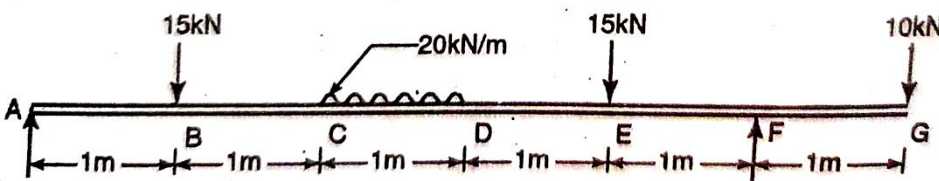


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : Civil Engineering Semester :III Subject Code & Name: BTCVES302 Mechanics of Solids Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Derive the relationship between modulus of elasticity and modulus of rigidity.	L2/CO1	6
B)	A specimen of steel 25mm diameter with a gauge length of 200mm is tested to destruction. It has an extension of 0.16mm under a load of 80 kN and the load at elastic limit is 160 kN. The maximum load is 180 kN. The total extension at fracture is 56mm and diameter at neck is 18mm. Calculate i) The stress at elastic limit ii) Young's Modulus iii) Percentage elongation iv) Ultimate tensile stress	L3/CO1	6
C)	Fig. shows a copper rod AB of length 600mm. When the temperature of the rod is 24 ⁰ C, the gap of BC is 0.48 mm. Determine i) Thermal stress at 40 ⁰ C ii) The temperature at which the gap just close iii) Stress and strain in the rod when its temperature is 88 ⁰ C Take E = 100 GPa and $\alpha = 16 \times 10^{-6}$ per ⁰ C	L3/CO1	6
 The diagram shows a vertical copper rod AB of length 600 mm. The top end A is fixed to a ceiling. The bottom end B is positioned 0.48 mm above a base labeled C. Arrows indicate the dimensions: 600 mm for the rod length and 0.48 mm for the gap between B and C.			

Q.2	Solve Any Two of the following.		12
A)	Draw SFD and BMD for a simply supported beam carrying point load at center	L3/CO2	6
B)	Draw Shear force and Bending moment diagram for the beam shown in figure 	L3/CO2	6
C)	Draw the SF and BM diagrams for the beam shown in fig. and mark the salient points. Find the point of contra flexure 	L3/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Write the assumptions made in the theory of pure bending	L1/CO3	6
B)	Derive and draw the shear stress distribution diagram for the rectangular cross section of the beam	L2/CO3	6
C)	A hollow shaft of 60mm outer diameter transmits 180 kW of power while rotating at a frequency of 25 hertz. Find the thickness of the shaft so that the shear stress does not exceed 60 N/mm ²	L3CO3	6
Q.4	Solve Any Two of the following.		12
A)	Define effective length of column, give its values for various end conditions of column and state the importance of effective length in Euler's formula	L1/CO3	6
B)	A 1.5m long column has a circular cross section of 50mm diameter, one of the end of the column is fixed in direction and position and the other is free. Taking factor of safety as 3. Calculate the safe load using i) Rankine's Formula with $f_c = 560\text{N/mm}^2$ and $a = 1/1600$ for Pinned end ii) Euler's Formula with E for column = $1.2 \times 10^5 \text{ N/mm}^2$.	L3CO4	6

C)	Evaluate the ratio of the buckling strengths of two circular columns one hollow and the other solid. Both the columns are made of the same material and have the same length, cross-sectional area and end conditions. The internal diameter of the hollow column is half of its external diameter.	L3/CO4	6
Q. 5	Solve Any Two of the following.		12
A)	What is Mohr's stress circle? How is it useful in the solution of stress analysis problems.	L2/CO1	6
B)	At a point in a piece of elastic material the normal stresses on two mutually perpendicular planes are 80 N/mm^2 tensile and 60 N/mm^2 compressive. These planes also carry shear stresses of 65 N/mm^2 . Determine the principal planes and the principal stresses	L3/CO1	6
C)	A solid circular shaft is subjected to a bending moment of 40 kN-m and a torque of 10 kN-m . Design the diameter of the shaft according to i) Maximum principal stress theory ii) Maximum shear stress theory Take $\mu = 0.25$, stress at elastic limit = 200 N/mm^2 and factor of safety = 2.	L3/CO1	6
	*** End ***		

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